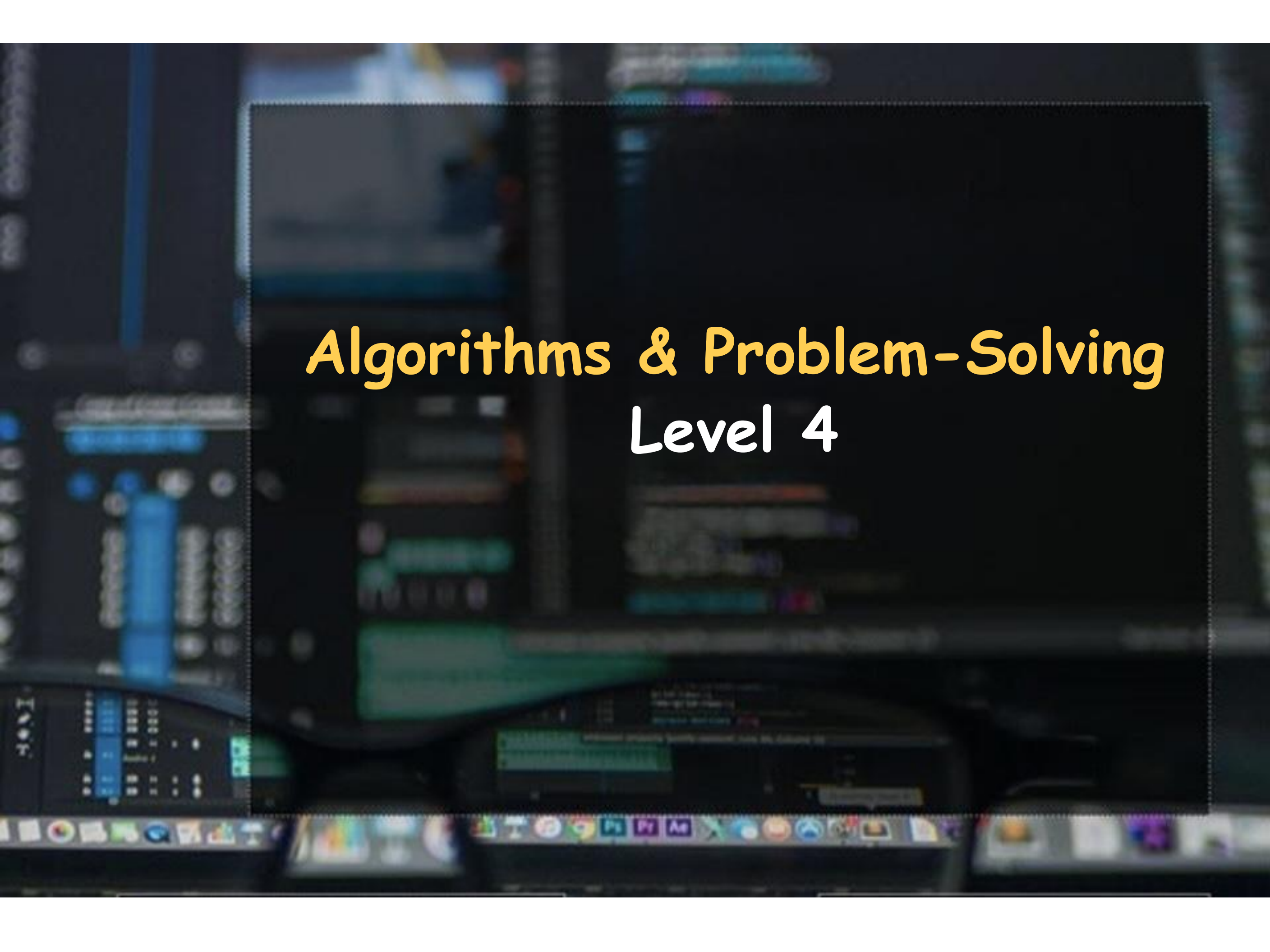
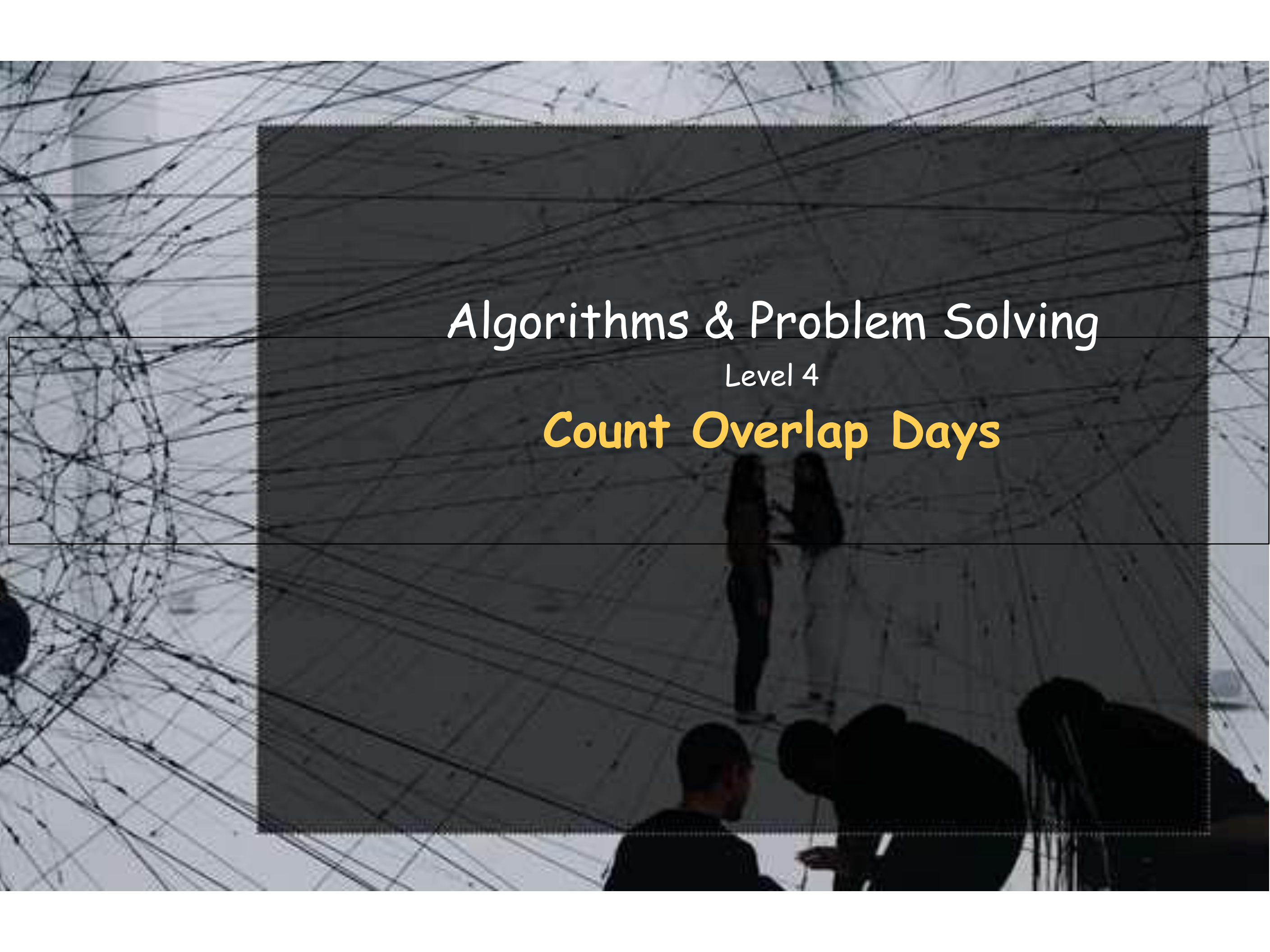




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Count Overlap Days

Definition

- sDate Struct: Represents a single date containing the day, month, and year.
- sPeriod Struct: Represents a time period containing:
 - StartDate: Start date
 - EndDate: End date
- enDateCompare Enum:
 - To determine the relationship between two dates:
- Before = -1 (Date before date)
- Equal = 0 (Dates are equal)
- After = 1 (Date after date)



Explanation:

- This program reads two time periods (each period has a start date and an end date) and then calculates the number of days shared between the two periods (overlapping days).

Procedures

- 1 Read the first period (beginning and end)
- 2 Read the second period (beginning and end)
- 3 Check if there is any overlap between the two periods
- 4 If there is an overlap, calculate the number of days that overlap between them
- 5 Print the number of overlapping days



Example:

Enter Period 1:

Start Date: 1/1/2022

End Date: 10/1/2022

Enter Period 2:

Start Date: 5/1/2022

End Date: 30/12/2050

Overlap Days Count Is: 5

Explanation:

- The program is reviewed for each day within the shorter period, and is verified if that day falls within the second period.
- Each overlapping day increases the number of shared days.



Comparison:

Status

No overlap

Partial overlap

One period is within another

Result

0 days

Only overlapping days are counted

The entire smaller period is counted



Note:

- If the periods are reversed (start date after end date), the results will be incorrect. Make sure the dates are in the correct order first.

Summary

- ✓ The program accurately calculates the number of days overlapping between two time periods through a structured daily comparison.
- ✓ It is used in applications related to vacations, bookings, or time period analysis.

Thank you 😊